Knife Gate Valve

I. This knife gate valve is suitable for shutting off the flow of fluid media such as slurry, sewage, and dust in pipeline systems.

II. Performance Specifications

Table Performance Specifications

Model	Nominal Diameter DN (mm)	Nominal Pressure PN (MPa)	Test Pressure (MPa)	Operating Temp.	Suitable Media
Z73H/F-16P	50 to 200	1.6	Strength: 2.4 Sealing: 1.8	≤300°C	Pulp slurry, sewage, coal slurry, ash, slag-water mixture

III. Structural Features

1. The raised gate seal face scrapes off deposits on the sealing surfaces and automatically removes debris.
2. The stainless steel gate prevents sealing leakage caused by corrosion.
3. The all-stainless steel construction prevents damage from corrosion.
4. The short face-to-face dimension saves material and installation space while effectively supporting pipeline strength.
5. Scientific packing gland design ensures safe, reliable, and durable upper sealing.
6. The triangular bracket saves raw material while ensuring the required mechanical properties.
7. Guide blocks on the valve body ensure correct gate motion; compression blocks ensure effective sealing.
8. The reinforced body rib design enhances valve body strength.
9. The stainless steel stem is durable; double-start threads enable faster opening and closing.
10. The basic structure of this valve is shown in Figure 1.

IV. Operating Principle

For manually operated valves, turn the handwheel by hand. The stem rises or descends, driving the gate plate up or down to open or close the valve. Rotating clockwise lowers the gate plate, closing the valve. Rotating counterclockwise raises the gate plate, opening the valve.

V. Storage, Maintenance, Installation, and Use

5.1 Valves shall be stored indoors in a dry and ventilated area, with both pipeline openings sealed.

5.2 Valves stored for extended periods shall be inspected regularly; debris shall be removed. Pay special attention to keeping the sealing faces clean to prevent damage.

5.3 Before installation, verify that the valve markings match the requirements.

5.4 Before installation, inspect the valve interior and sealing faces; if any contaminants are present, wipe clean with a soft cloth.

VI. Possible Faults, Causes, and Resolutions — See Table

Table Possible Faults, Causes, and Resolutions

Possible Fault	Cause	Resolution
Packing leakage	1. Packing gland not tightened 2. Packing failed due to prolonged use or improper storage	1. Evenly tighten the nuts to compress the packing 2. Replace the packing
Sealing face leakage	1. Debris on sealing faces 2. Sealing face damaged	1. Remove all debris thoroughly 2. Re-machine and resurface,

Possible Fault	Cause	Resolution
		or replace
Leakage at body-cover joint	1. Joint bolts unevenly tightened 2. Flange sealing face damaged	1. Tighten evenly 2. Re-face the flange
Handle difficult to turn / gate fails to open or close	1. Packing over-compressed 2. Stem nut damaged 3. Stem nut threads severely worn or broken 4. Stem bent	1. Slightly loosen packing gland nuts 2. Disassemble and repair threads, clear debris 3. Replace stem nut 4. Straighten the stem